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Advantages, manufacturing processes and challenges of FinFET technology

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Abstract. As an emerging cutting-edge integrated circuit technology, Fin Field Effect Transistor (FinFET) has not only successfully broken through the limitations of the 22nm node since its development in 1998, but it still maintains considerable vitality and is developing into smaller sizes. As an improvement over traditional MOSFET technology, FinFET offers many technical advantages, which practically solve the challenges encountered by traditional MOSFET. The FinFET process technology is incompatible with planar MOSFET process technology. Moreover, since the development of Finfet technology, there have been many technical challenges. The first half of this article mainly introduces the differences and advantages between FinFET and traditional Metal-Oxide-Semiconductor Field Effect Transistor (MOSFET), and briefly outlines its manufacturing process. The second half explains the challenges and hopes on the development path of FinFET technology. I believe this article can provide a reference for integrated circuit designers when understanding FinFET technology, and provide guidance and reference for beginners.

Keywords: FinFET, Advantages, Manufacturing processes, Challenges.

1. Introduction

Since its invention in the 1950s, integrated circuit (IC) technology has been the focus of modern science and technology. Under the guidance of Moore's Law, IC technology develops rapidly, and chip performance continues to exceed the limits of human technology. Before the 22nm node, when the channel length was shrinking, the IC industry had been increasing capacitance per unit area gate-oxide by thinning gate oxide thickness, suppressing the impact of drain voltage on the channel area by reducing channel width, using an light doped drain structure and reducing the power supply voltage to suppress the strong electric field at the channel drain end effect; use Halo structure and reverse doping distribution to suppress drain-induced barrier lowering and subsurface punch-through; introduce strain technology to improve the mobility of channel carriers; use High-K+ metal gate to suppress tunneling of thin gate dielectrics. These technologies allow the MOSFET to still maintain the long channel electrical characteristics when the channel length is shortened from 10 microns to 28 nanometers.

At the 22nm node, it is becoming more and more difficult to reduce the dimensions of gate oxide thickness and channel width. Therefore, in order to break through the limitations of the 22nm node, new device structures have emerged. In 1998, in the University of California, Berkeley, Professor Chenming Hu led his team to invent Fin Field Effect Transistor (FinFET).



The first half of this article mainly introduces the differences and advantages between FinFET and traditional MOS devices, especially in terms of short channel effects, and briefly outlines its manufacturing process. The second half explains the challenges and hopes on the development path of FinFET technology. It is believed that the article can provide a reference for integrated circuit designers when understanding FinFET technology, and provide guidance and reference for beginners.

1.1. Introduction to FinFET

FinFET's full name is Fin Field-Effect Transistor, which is an advanced kind of complementary metal oxide semiconductor transistor. It is named FinFET because the similarity between its shape and the fin of a fish [1].

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When it comes to the 22nm node, it is becoming more and more difficult to reduce the dimensions of tox. Even if these dimensions could be continued to be reduced, this would only give the gate voltage a strong control over the silicon surface. At the subsurface, the leakage voltage controls the potential more strongly than the gate, resulting in subsurface leakage current and making the off-state current larger. Many challenges encountered by traditional MOSFETs are also difficult to solve. For example, small size effect, roll-off of threshold voltage, mobility degradation and velocity saturation effect (VSE), hot-carrier effect (HCE). At extremely short channel lengths, carrier scattering can be ignored, and the channel exhibits ballistic transport. In order to achieve breakthroughs in chip performance, cost and power consumption, the size of transistors has been shrinking since they were developed in the last century. In terms of performance, reducing the feature size of the device can improve the performance of the device. In terms of cost, shrinking device size allows more circuits to be fabricated on a wafer, making each circuit less expensive. In terms of power consumption, reducing the area of the device will reduce the gate capacitance and source-drain PN junction capacitance, thereby reducing the dynamic power consumption of the device. In practice, people find that the performance of the device begins to deviate from the characteristics predicted by the long channel theory when the channel length declines, and a series of undesirable electrical characteristics appear. These deviations from long channel characteristics are collectively called small size effects. The main small size effects are: threshold voltage is related to the geometric size of the channel and drain voltage, sub-threshold swing increases, drain in sub-threshold region and saturation region The current is related to the drain voltage, and the leakage current in the saturation region deviates from the square rate relationship as the gate voltage changes. Because most deviations are related to the reduction of channel length, the small size effect is also called the Short-Channel Effect, or SCE for short [2].

As the channel length shrinks, the depletion regions of the source and drain PN junctions have a significant impact on the potential distribution in the channel region, making the channel electric field become two-dimensional, and the gradually varying channel is approximately in most areas of the channel region. Invalid. Moreover, the electric field along the channel direction is enhanced, causing changes in the carrier transport characteristics in the channel, and even generating hot carriers at the drain end of the channel, causing strong field effects such as impact ionization and hot carrier injection. Because of these small size effects, shrinking transistor sizes is becoming increasingly difficult. In order to continue Moore's Law, FinFET came into being.

In 1998, Professor Chenming Hu led a research team at the University of California to study how to expand Complementary Metal Oxide Semiconductor process technology to the 25nm field. Based on

the 3-dimensional structure of MOS transistors and double-gate MOSFET structures, professor Chenming Hu further developed a self-aligned double-gate MOSFET structure. The transistor got its name as FinFET transistor for its shape is similar to a shark fin.

In the traditional transistor structure, the gate is only able to control the off and on of the circuit on single side of the gate, and it is called a planar structure. When it comes to the FinFET architecture, its gate forms a fork-shaped 3D structure, which can control the on and off on both sides of the circuit. The channel is surrounded by the gate from three sides. A three-dimensional structure is developed instead of a planar device to enhance the control capability to the gate. After applying a voltage, an inversion layer conductive channel is formed under the gate oxide layer like the traditional transistor to control the flow of carriers, thereby achieving its switching function.

2. Differences and advantages of finfet

As an improvement over traditional MOSFET technology, FinFET offers the following advantages. Firstly, it is generally lightly doped in the channel of FinFET, and the scattering effect of discrete doped atoms can get avoided. Its carrier mobility is greatly improved compared to heavily doped traditional devices. Secondly, in narrow fins, volume inversion as well as very low channel doping can cause saturation velocity and better low field carrier mobility. Additionally, ballistic transport in sub-20nm channel lengths and strain engineering further improve carrier mobility. It is crucial to obtaining consistent improvement in drive current through subsequent technology nodes.

Moreover, the FinFET mechanism enhances the control area of the gate to the channel and improves the gate control capability, which are able to reduce sub-threshold leakage current and suppress the short channel effect. The sub-threshold swing is a performance index that measures the mutual conversion rate between the on and off states of a transistor. It shows the change in gate voltage required for a tenfold change in source and drain current. The smaller swing means the faster the on-off rate. At present, the limit value of the swing is about 60mV/dec at room temperature, and it is difficult to reduce as the device size shrinks. However, in FinFET, as the silicon Fin thickness decreases, the sub-threshold slope also decreases and approaches the ideal value of 60mV/dec. This is because as the thickness of the silicon Fin decreases, the gate's ability to control the channel will gradually increase, thereby inhibiting the short channel effect.

Because of the improvement of gate control capabilities and the suppression of short channel effects, it is definitely possible for FinFET devices to apply thicker gate oxides than traditional devices, thereby reducing gate leakage current. Moreover, Stronger capability of gate control allows the transistor's gate length to be significantly shortened, thereby further reducing the area.

Figures (a)–(c) have been dedicated to the discussion of FinFET short-channel characteristics in Figure 1. A noteworthy point can be extracted from Figure (a): threshold voltage roll-off is noticeably improved as fin width is decreased. Figure (b) and (c) provide us with a different angle, recording the values of saturation subthreshold swing and DIBL, at fin widths ranging from 20 to 150 nm, gate lengths from 10 to 42 nm, for drain voltage $|V_d|=1\text{V}$ and current $|I_d|=100\text{nA}/\mu\text{m}$. It is worth noting that when the ratio of gate length to fin width exceeds 1.5, DIBL and subthreshold slope are both below 100 mV/dec and 0.1 V/V, respectively, for NMOS and PMOS devices. PMOS devices, on the other hand, display more severe short-channel effects for phosphorus diffusivity is lower than boron diffusivity, which causes a lower L_{eff} after rapid thermal annealing for source/drain activation. The added bonus of FinFETs is the extremely low gate-induced drain leakage current (GIDL) they display, as Figure (d) clearly indicates [3].

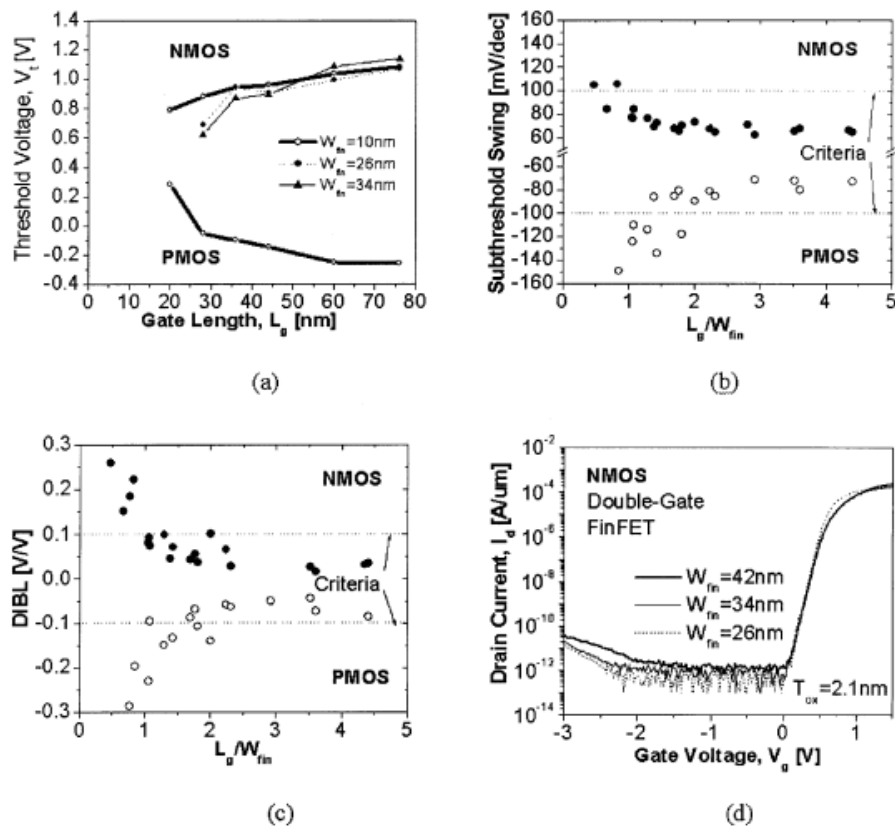


Figure 1. GIDL and FinFET short-channel effects [1]. (a) Threshold voltage rolloff versus L . (b) Subthreshold swing versus $L = W$. (c) DIBL versus $L = W$. (d) GIDL current.

3. FinFET manufacturing process

The FinFET process technology is incompatible with planar MOSFET process technology. The advanced FinFET process adopts a three-dimensional structure. HKMG technology and strained silicon technology are included as well, while the back-end process still utilizes the Damascus copper process structure. The difficulty of the FinFET process lies in the formation of the Fin shape, with the Fin size being approximately 0.67 times the minimum gate length. When it comes to 22nm process technology, the Fin width is 14.67nm, and it is far smaller than the smallest one that can be manufactured by the most precise immersion lithography machine. The active area of the Fin is not formed using photolithography, but rather by Self-Aligned Double Patterning (SADP) process technology. Only one photolithography step is required, followed by an auxiliary process similar to gate sidewall formation [4].

Regarding the SADP process, first deposit an auxiliary layer of polysilicon or Si_3N_4 , then form a gate-like structure through photolithography and etching, usually called a mandrel, and then deposit a layer of silicon oxide as For the hard mask, the width W_{fin} of Fin can be controlled by controlling the thickness of silicon oxide. Dry etching is used to form a shape similar to the gate sidewall. After removing the auxiliary layer, the remaining shape is a hard mask to form an ultra-thin Fin. Membrane, and then use dry etching to form ultra-thin Fin. The following is the specific process of FinFET process.

3.1. Fin Formation

The FinFET process's difficulty lies in the formation of the Fin. Because the size of the Fin is too small, it is not formed directly through photolithography, but through process technologies such as SADP (Self-Aligned Double Patterning).

A layer of SiO_2 film used as an etching barrier is formed on the Si wafer, amorphous Si is grown and deposited on it, and then photolithography and dry etching are performed, and the photoresist is ashed to form an amorphous Si intermediate material. SiN_x or SiO_2 is deposited on the amorphous Si intermediate material, and the sidewalls are formed through a dry etching process. After the intermediate material formed by the amorphous Si is removed, a hard mask of the Fin structure is formed. Using dry etching, a Fin structure is formed on the Si wafer [5].

It begins with the formation of N well and P well. There are two methods of forming wells in FinFET fabrication: (1) at the beginning of the process, and (2) after the fins are patterned. That is the Well first or Fin first mentioned earlier. The steps introduced today are Fin first. After Fin is formed, Si is protected by generating SiO_2 and Si_3N_4 layers, and then N well and P well are formed through ion implantation.

Secondly, it is the STI Filling. The Si wafer in the Fin structure is filled with SiO_2 using a high filling ratio process. After an annealing, the CMP process and the SiO_2 Etch Back process are used to etch and remove part of the SiO_2 to form STI [6].

The next step is the formation of Gate. The Gate here is equivalent to the Dummy Gate. In the subsequent preparation process, the polysilicon gate will be removed, high- κ insulating dielectric and metal gates will be deposited, and multi-layer metal gates will be etched.

Moreover, there is the Strained Silicon Process. In the same plane process, strained silicon technology will also be used in FinFET. By embedding SiGe and SiC strained materials in the epitaxial layer, tensile stress is formed on NMOS and compressive stress is formed on PMOS, thus increasing the FinFET driving capability.

4. Challenges and expectations of finfet

Since the development of FinFET technology, there have been many technical challenges. First, width quantification is a problem. When designing FinFET devices, FinFET components are placed as regular structures. The designer can change the width of the planar transistor to achieve higher performance, but cannot change the height or width of the fin. Therefore, if the device performance cannot be improved, the number of fins can only be increased.

Meanwhile, parasitic capacitance has always been the trouble. Compared to planar structures, the disadvantage of FinFET structures is they have more parasitic capacitance. The additional overlap area on the front and back gates increases the capacitance. Changes in fin height help reduce the capacitance, that is, increasing fin height and reducing fin spacing help reduce the parasitic capacitance of FinFETs. Parasitic capacitance's analytical modeling can be accomplished through conformal mapping techniques.

What's more, doping concentration of FinFET has to be fixed. Undoped channels are preferred for FinFET technology. However, light doping is recommended for better control of leakage current. The doping concentration of the source/drain regions demands higher dopants, thereby increasing the series resistance. It can damage the geometry of the device's fins. In order to handle this issue, in-situ epitaxial growth is performed on the source and drain regions, with or without removal of the fins of this structure.

Process integration challenges cannot be ignored as well. Process integration challenges generally include the following: a. Precise and uniform fin patterning; b. 3D gate and sidewall patterning; c. Uniform bonding in the fin; d. Stress engineering [7].

FinFET reliability should be considered. Lower lateral fields in the transistor can be realized through fully depleted FinFET operation, which can enhance the reliability of NMOS dielectric breakdown. Instabilities in threshold voltage have been discovered in the transition from 32 nm planar technology to 22 nm FinFET technology as well. There appears to be no change in the reliability of PMOS in FinFET devices. Moreover, FinFET will cause a sudden increase in local transistor current density, which means that the reliability of metal electromigration of signal lines and power ground networks will be affected even more. 3D devices have higher parasitics, tighter interconnect spacing, higher negative bias temperature instability (NBTI) and more complex gates Stacked topology, and complex time-dependent dielectric breakdown (TDDB) reliability challenges.

Last but not the least, etching challenges serve as the problem. The channel of the three-dimensional structure of FinFET is surrounded by the gate, which causes the gate to be etched to face more challenges

than the planar structure. First, the top of Fin is a relatively small structure and is more easily damaged by plasma. After etching the Fin and removing the top hard mask, it is generally necessary to grow a thin oxide layer through thermal oxidation to repair the surface damage of the Fin caused by plasma etching; at the same time, the sharp corners on the top of the Fin are rounded to avoid sharp corners in the device. Reliability problems caused by Corner Effect [8]. The selective and residue-free etching process is challenging in the manufacturing of FinFET devices. It is necessary to avoid source and drain damage as much as possible while ensuring a steep etching topography, and to ensure that there is no polycrystalline residue at the corner where the gate and Fin meet.

With finfet technology developing, nowadays there are expectations to be fulfilled. As the device size further shrinks, the requirements for gate control capabilities will further increase, and FinFET may evolve into the shape of a columnar FET. Compared with FinFET, whose gate surrounds the channel on three sides, its gate surrounds the entire channel, so it has better characteristics than FinFET. And because there are no sharp corners, there is no corner effect. A single device has a smaller size and can easily achieve higher integration to further improve integrated circuit performance [9].

FinFET structures, on the other hand, are less efficient at dissipating heat because heat can easily accumulate on the fins. These problems may lead to a new class of design rules - Thermal Design, which involves modifying the device structure and replacing existing silicon materials with new materials [10].

5. Conclusion

Under the guidance of Moore's Law, IC technology develops rapidly, and chip performance continues to exceed the limits of human technology. As an advancement in traditional MOSFET technology, FinFETs possess certain benefits and unique challenges. When compared to planar devices, the FinFET structure provides better channel control capabilities, improved sub-threshold slope, lower leakage current, reduced gate delay, and stronger current drive capability. Additionally, the FinFET design increases the gate-to-channel control area, greatly enhancing gate control effectiveness. Owing to the suppression of short-channel effects and improved gate control, it is practical for FinFET devices to apply thicker gate oxides than traditional devices, which can reduce gate leakage current. FinFETs show promise for application in the 22nm technology generation and beyond. The channel of FinFETs is generally lightly doped, avoiding the scattering effects of individual dopant atoms. However, issues such as parasitic capacitance and doping concentration still pose challenges in the development of FinFET technology. Currently, in terms of gate control capabilities and heat dissipation efficiency in FinFET technology, there is still room for improvement.

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